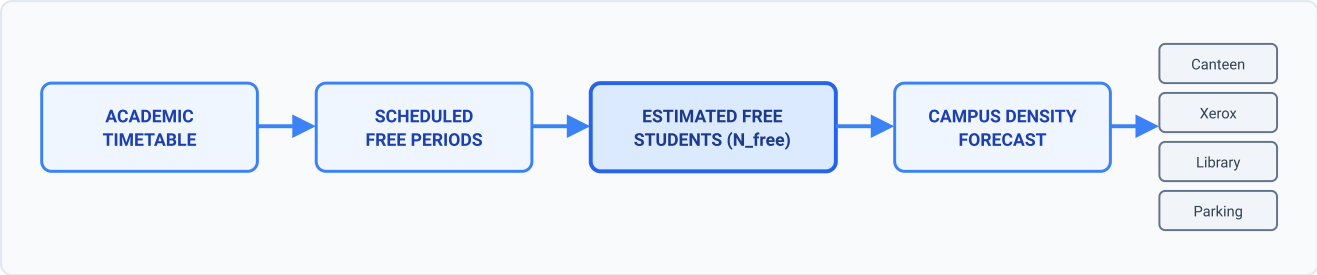


**Central Data Science Hypothesis:** Academic schedules, section capacities, and historical class attendance patterns can be aggregated to forecast real-time student availability, serving as a leading indicator to predict demand, queue lengths, and resource bottlenecks across shared campus facilities.

## 1. SYSTEM CONCEPTUAL FRAMEWORK Leading vs. Lagging Indicators

Traditional facility management relies on *lagging indicators*—reactively detecting crowds after they form via Wi-Fi logs or overhead cameras. CampusPulse introduces a *leading indicator* paradigm by leveraging academic timetables to model movement vectors **15 to 45 minutes prior** to crowd formation.



## 2. PRIVACY-PRESERVING DATA ARCHITECTURE Aggregated Section Level

CampusPulse operates strictly at the **Division / Section level** (e.g., TY-CS-N), completely eliminating personal data handling, individual student IDs, or privacy tracking compliance hurdles.

Table 1.1: Academic Schedule Entity Schema (‘Timetable’)

| DIVISION ID | DAY | TIME SLOT     | SCHEDULED SUBJECT       | ROOM     | IS LECTURE ACTIVE?  |
|-------------|-----|---------------|-------------------------|----------|---------------------|
| TY-CS-N     | Mon | 09:00 - 10:00 | Data Structures (DS)    | Room 201 | Yes                 |
| TY-CS-N     | Mon | 10:00 - 11:00 | Computer Networks (CN)  | Room 201 | Yes                 |
| TY-CS-N     | Mon | 11:00 - 12:00 | Operating Systems (OS)  | Room 201 | Yes                 |
| TY-CS-N     | Mon | 12:00 - 13:00 | FREE PERIOD             | —        | No (Scheduled Free) |
| TY-CS-N     | Mon | 13:00 - 14:00 | Database Systems (DBMS) | Room 201 | Yes                 |

3. MATHEMATICAL FORMULATION & FEATURE ENGINEERING

Population Modeling

A scheduled free period does not guarantee 100% presence due to absenteeism, off-campus mobility, or library study. We model actual available population using an **Attendance-Weighted Presence Factor**.

Free Student Population Formula:

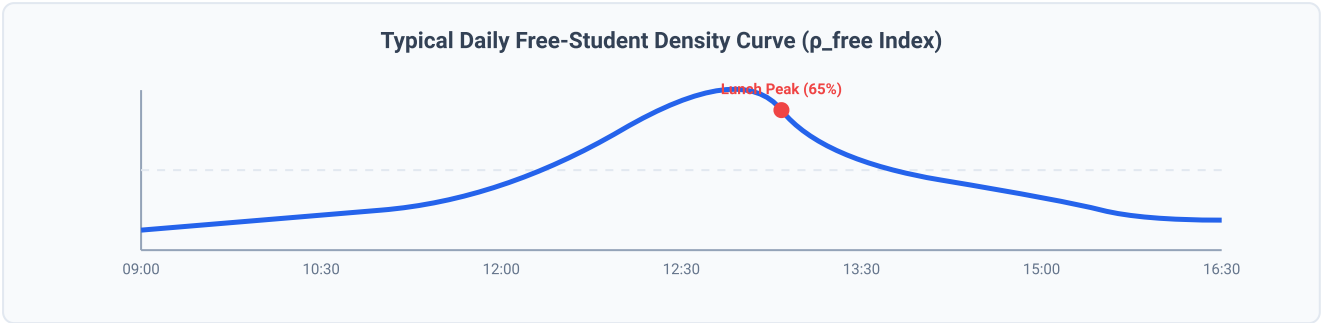
$$N_{free}(t) = \sum_{d \in D_{free}(t)} ( N_{total}(d) \times \alpha(d, day, t) )$$

Where  $D_{free}(t)$  is the set of unassigned divisions at time slot  $t$ ,  $N_{total}(d)$  is total division strength, and  $\alpha(d, day, t)$  is the historical campus presence coefficient ( $0.0 \leq \alpha \leq 1.0$ ).

4. DERIVED FEATURE: CAMPUS FREE-STUDENT DENSITY

Feature Vector

By computing the ratio of free students against total enrolled campus population, we generate a continuous feature vector representing instantaneous campus pressure ( $\rho_{free}$ ).



5. MULTI-MODEL MACHINE LEARNING PIPELINE

Tier-1 & Tier-2 Architecture

|                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tier-1: Population Estimator</b><br><b>Model 1 – Free Student Model</b><br><br><b>Inputs:</b> Timetable slot, active division schedules, historical attendance matrix, exam flags.<br><b>Output:</b> Estimated free student count ( $N_{free}$ ). | <b>Tier-2: Facility Predictive Models</b><br><b>Model 2 – Canteen Queue Predictor</b><br><br><b>Inputs:</b> $N_{free}$ , live counter status, active order rate.<br><b>Output:</b> Predicted waiting time in minutes. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 2.1: Secondary Facility Models & Target Outputs

| FACILITY MODEL               | PRIMARY MODEL INPUTS                                  | TARGET OUTPUT METRIC          | ALGORITHMIC APPROACH          |
|------------------------------|-------------------------------------------------------|-------------------------------|-------------------------------|
| <b>Model 3: Xerox Center</b> | Pending print queue, active hardware, $N_{free}$      | Estimated Service Delay (min) | Random Forest Regression      |
| <b>Model 4: Library</b>      | Current seating fill, exam proximity flag, $N_{free}$ | Predicted Occupancy Rate (%)  | Ridge / Polynomial Regression |
| <b>Model 5: Parking Bay</b>  | Vehicle entry/exit rate, current occupied bays        | Available Spaces Forecast     | Time-Series LSTM / Prophet    |

6. CENTRAL INTELLIGENCE DASHBOARD INTERFACE

Real-Time Operational Wireframe

The operational interface provides facility managers with actionable forecasts 30 minutes ahead of peak queue formation.

CAMPUSPULSE // CENTRAL INTELLIGENCE ENGINE

STATUS: LIVE PROJECTION

TIMETABLE SNAPSHOT: 12:15 PM | ESTIMATED FREE POPULATION: 205 STUDENTS (HIGH DENSITY)

CANTEEN FACILITY

CRITICAL DEMAND ALERT

Current Queue: 14 people  
30-Min Forecast: 18 min wait time  
Rec: Open Counter 3 & 4 immediately.

XEROX & PRINT STATION

MODERATE DEMAND

Current Queue: 4 people  
30-Min Forecast: 7 min wait time  
Status: Operational.

MAIN LIBRARY

CAPACITY FILLING

Seats Free: 42 of 250  
30-Min Forecast: 82% Occupancy  
Status: Moderate study inflow.

STUDENT PARKING

STABLE AVAILABILITY

Available Bays: 18 slots  
30-Min Forecast: 12 slots available  
Status: Vehicle inflow slowing.

7. COMPARATIVE STRATEGIC ADVANTAGES

Platform Evaluation

Reactive Hardware Solutions

- Requires expensive IoT sensors & cameras.
- Only alerts after queues have formed.
- High maintenance and hardware overhead.
- Potential privacy/surveillance concerns.

CampusPulse Predictive Engine

- Uses existing academic timetable data.
- Forecasts bottlenecks 30+ minutes in advance.
- Zero hardware deployment costs.
- 100% privacy-preserving division-level analytics.

8. IMPLEMENTATION ROADMAP

Deployment Strategy

- Phase 1: Timetable Ingestion** — Integrate master schedule database and division counts.
- Phase 2: Baseline Model Calibration** — Train free-student estimation models on historical attendance trends.
- Phase 3: Facility Connector Integration** — Connect live queue telemetry from canteen POS and library gates.
- Phase 4: Predictive Dashboard Rollout** — Deploy real-time intelligence feeds to administrative units.